

Lecture 4

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- Vector spaces and linear maps
- Norms and inner products
- Convergence in $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$

- Vector space
- Linear map/transformation
- Norm
- Inner product
- Cauchy-Schwarz inequality
- Submultiplicativity
- Matrix norms
- Vector norms
- Convergence in norm

Linear algebra refresher

Before studying multivariate optimization, we need linear algebra and multivariable calculus.

In this course, we will be primarily concerned with two specific vector spaces:

Vectors: $\mathbb{R}^n$

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\}$$

Matrices: $\mathbb{R}^{m \times n}$

$$\mathbb{R}^{m \times n} = \left\{ \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} : x_{ij} \in \mathbb{R}, 1 \leq i \leq m, 1 \leq j \leq n \right\}$$

We will also need subspaces like

$$\mathbb{S}^n := \{X \in \mathbb{R}^{n \times n} : X^T = X\} \quad (\text{set of all symmetric matrices})$$

Reminder: a **subspace** is a vector space that is contained within another vector space.

It inherits the vector space structure from the larger space (such as the operations of vector addition and scalar multiplication) and must be closed under these operations.

We can combine two vector spaces to form a new vector space.

Every element of a product space is a pair of elements, one from each of the original spaces.

For example, the product of $\mathbb{R}^3$ and $\mathbb{R}^2$ is the set of all pairs of 3×1 vectors and 2×1 vectors.

Some examples of product spaces:

$$\mathbb{R}^{m \times n} \times \mathbb{R}^{n \times p}, \quad \mathbb{R}^{m \times n} \times \mathbb{R}^n, \quad \mathbb{R}^m \times \mathbb{R}^{m \times n} \times \mathbb{R}^n, \quad \mathbb{S}^n \times \mathbb{R}^n, \dots$$

Note that $\mathbb{R}^n = \mathbb{R}^{n \times 1}$ and so everything we say about $\mathbb{R}^{m \times n}$ specializes to $\mathbb{R}^n$.

You will also need to know about linear maps (a.k.a. linear transformations or linear operators), i.e., a function $T : \mathbb{V} \rightarrow \mathbb{W}$ that satisfies

$$T(a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2) = a_1 T(\mathbf{v}_1) + a_2 T(\mathbf{v}_2).$$

The set of all linear maps between two vector spaces $\mathbb{V}$ and $\mathbb{W}$ will be denoted by

$$\mathcal{L}(\mathbb{V}, \mathbb{W}) := \{T : \mathbb{V} \rightarrow \mathbb{W} \mid T \text{ is linear} \}.$$

This is itself a vector space.

Exercise

Which of the following maps are linear? Find the rank of those linear maps.

1. $T_1 : \mathbb{R}^{m \times n} \times \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{m \times p}, (X, Y) \mapsto XY$.
2. $T_2 : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}, X \mapsto AXB$ where $A \in \mathbb{R}^{p \times m}, B \in \mathbb{R}^{n \times q}$ are fixed constant matrices.
3. $T_3 : \mathbb{R}^{m \times q} \times \mathbb{R}^{p \times n} \rightarrow \mathbb{R}^{p \times q}, (X, Y) \mapsto AX + YB$ where $A \in \mathbb{R}^{p \times m}, B \in \mathbb{R}^{n \times q}$ are fixed constant matrices.
4. $T_4 : \mathbb{R}^m \times \mathbb{R}^{m \times n} \times \mathbb{R}^n \rightarrow \mathbb{R}, (\mathbf{x}, Y, \mathbf{z}) \mapsto \mathbf{x}^\top Y \mathbf{z}$.
5. $T_5 : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}, (\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x}^\top A \mathbf{y}$ where $A \in \mathbb{R}^{m \times n}$ is a fixed constant matrix.
6. $T_6 : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}, X \mapsto \mathbf{a}^\top X \mathbf{b}$ where $\mathbf{a} \in \mathbb{R}^m$ and $\mathbf{b} \in \mathbb{R}^n$ are fixed constant vectors.

Norms and Inner Products

Definition (Norm)

Let $\mathbb{V}$ be $\mathbb{R}^n$ or $\mathbb{R}^{m \times n}$. A norm $\|\cdot\| : \mathbb{V} \rightarrow \mathbb{R}$ is a function satisfying

1. $\|v\| \geq 0$ for all $v \in \mathbb{V}$;
2. $\|v\| = 0$ iff $v = 0$;
3. $\|\lambda v\| = |\lambda| \|v\|$ for all $\lambda \in \mathbb{R}$ and $v \in \mathbb{V}$;
4. $\|v + w\| \leq \|v\| + \|w\|$ for all $v, w \in \mathbb{V}$.

The most common norms, and in fact the only norms we will use, in this class are:

Vector 2-norm: $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ with

$$\|\mathbf{x}\| = \left[\sum_{i=1}^n x_i^2 \right]^{1/2}$$

(Usually written $\|\cdot\|_2$ but we drop the subscript when clear from context)

2. Frobenius norm: $\|\cdot\| : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ with

$$\|X\| = \left[\sum_{i=1}^m \sum_{j=1}^n x_{ij}^2 \right]^{1/2}$$

(Usually written $\|\cdot\|_F$ but we drop the subscript when clear from context)

Frobenius and 2-norm agree when m or n is 1

If $m = 1$ or $n = 1$, then the matrix is a row or column vector, respectively.

In this case, the Frobenius norm and 2-norm agree.

Suppose $X \in \mathbb{R}^{1 \times n}$. Then

$$\|X\|_F = \left[\sum_{i=1}^1 \sum_{j=1}^n x_{ij}^2 \right]^{1/2} = \left[\sum_{j=1}^n x_{1j}^2 \right]^{1/2} = \|X\|_2$$

In this case, no ambiguity in using the same notation $\|\cdot\|$ for both. We will refer to $\|\cdot\|$ as the “Euclidean norm”.

There are other matrix and vector norms but they not used in this class:

- Vector 1-norm: $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$. Measures distance measured along grid lines.
- Vector ∞ -norm: $\|\mathbf{x}\|_\infty = \max \{|x_1|, \dots, |x_n|\}$. Measures maximum distance measured along any of the axes.
- Matrix 2-norm or spectral norm: $\|X\|_2 = \sup_{\mathbf{y} \neq 0} \frac{\|X\mathbf{y}\|_2}{\|\mathbf{y}\|_2}$. Measures maximum stretching of a unit sphere.
- Matrix 1-norm or maximum column sum:
 $\|X\|_1 = \sup_{\mathbf{y} \neq 0} \frac{\|X\mathbf{y}\|_1}{\|\mathbf{y}\|_1} = \max_{j=1, \dots, n} \sum_{i=1}^m |x_{ij}|$. Measures maximum distance measured along the direction of any of the columns.
- Matrix ∞ -norm or maximum row sum:
 $\|X\|_\infty = \sup_{\mathbf{y} \neq 0} \frac{\|X\mathbf{y}\|_\infty}{\|\mathbf{y}\|_\infty} = \max_{i=1, \dots, m} \sum_{j=1}^n |x_{ij}|$. Measures maximum distance measured along the direction of any of the rows.

The operator $\text{vec} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{mn}$ takes the n column vectors of an $m \times n$ matrix and concatenate them into a long vector with mn entries, e.g.,

$$\text{vec} \left(\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} \right) = \begin{bmatrix} a_1 \\ a_2 \\ b_1 \\ b_2 \\ c_1 \\ c_2 \end{bmatrix}.$$

This gives us one more norm:

Matrix Hölder p -norm: $\|X\|_{H,p} = \|\text{vec}(X)\|_p$, most commonly, $p = 1, 2, \infty$.

Note that $\|X\|_{H,2} = \|X\|_F$ is the Frobenius norm we introduced earlier.

Most (but not all) matrix norms have an additional submultiplicativity inequality:

$$\|XY\| \leq \|X\|\|Y\|$$

for any $X \in \mathbb{R}^{m \times n}$ and $Y \in \mathbb{R}^{n \times p}$.

Norms with submultiplicativity include $\|\cdot\|_F$ and $\|\cdot\|_p$ for $p = 1, 2, \infty$, but not $\|\cdot\|_{H,p}$ for $p = 1, \infty$.

Definition (Inner product)

An **inner product** is any function $\langle \cdot, \cdot \rangle : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{R}$ satisfying

1. Bilinearity: $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$ for all $u, v, w \in \mathbb{V}$ and $\alpha, \beta \in \mathbb{R}$;
2. Symmetry: $\langle u, v \rangle = \langle v, u \rangle$ for all $u, v \in \mathbb{V}$;
3. Positive definite: $\langle v, v \rangle \geq 0$ for all $0 \neq v \in \mathbb{V}$;
4. $\langle v, v \rangle = 0$ iff $v = 0$.

The most common inner products are:

1. The dot product on $\mathbb{R}^n$: $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$.
2. The trace inner product on $\mathbb{R}^{m \times n}$: $\langle X, Y \rangle = \text{tr}(X^\top Y)$.

As a reminder, if we have two vectors $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$, then their dot product is given by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

The function defined by $\|v\| = \sqrt{\langle v, v \rangle}$ for $v \in \mathbb{V}$ will always be a norm on $\mathbb{V}$.

This is called the **norm induced by the inner product**, or just the **induced norm**.

The norm induced by the dot product in $\mathbb{R}^n$ is just the vector 2-norm $\|\cdot\|_2$.

The norm induced by the trace inner product on $\mathbb{R}^{m \times n}$ is the Frobenius norm $\|\cdot\|_F$.

In fact, the biggest reason $\|\cdot\|_2$ and $\|\cdot\|_F$ are important and ubiquitous is because they are induced by inner products - none of the other matrix and vector norms we discussed above are induced by inner products.

There is an important relationship between the inner product of two vectors and the norms of the two vectors. This is the **Cauchy-Schwarz inequality**.

Theorem (Cauchy-Schwarz inequality)

$$|\langle v, w \rangle| \leq \|v\| \|w\|$$

for all $v, w \in \mathbb{V}$.

The Cauchy-Schwarz inequality tells us that the length of the projection of one vector onto another is at most the length of the original vector.

If we project a vector v onto another vector w , the length of the projection is given by

$$\frac{|\langle v, w \rangle|}{\|w\|} \leq \|v\| \quad \text{from Cauchy-Schwarz inequality}$$

Cauchy-Schwarz inequality for vectors and matrices

Using the dot product for vectors and the trace inner product for matrices, we can write the Cauchy-Schwarz inequality for vectors as

$$|\mathbf{x}^\top \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|;$$

and the Cauchy-Schwarz inequality for matrices as:

$$|\mathrm{tr}(X^\top Y)| \leq \|X\| \|Y\|.$$

We will only need three inequalities in this course! We have already seen two of them:

- Cauchy-Schwarz inequality $|\langle v, w \rangle| \leq \|v\| \|w\|$
- Submultiplicativity inequality $\|XY\| \leq \|X\| \|Y\|$ (for some norms)

The third one is the **triangle inequality**:

$$\|X + Y\| \leq \|X\| + \|Y\|$$

Exercise

Prove these three inequalities for $\mathbb{R}^{m \times n}$:

$$\|X + Y\| \leq \|X\| + \|Y\|, \quad |\operatorname{tr}(X^\top Y)| \leq \|X\| \|Y\|, \quad \|XY\| \leq \|X\| \|Y\|$$

Note that you get the corresponding inequalities on $\mathbb{R}^m$ for free by setting $n = 1$.

Subspaces and product spaces

Two of the simplest ways to obtain new vector spaces out of old ones are to take **subspaces** and **products** of vector spaces.

Subspaces are just subsets of a vector space that are themselves vector spaces. For example, the set of all symmetric matrices is a subspace of $\mathbb{R}^{n \times n}$.

Products of vector spaces are just the sets of all possible ordered pairs of vectors from the two spaces. For example, the set of all ordered pairs of $m \times n$ and $p \times q$ matrices is a product space $\mathbb{R}^{m \times n} \times \mathbb{R}^{p \times q}$.

Subspaces inherit norms and inner products

Any subspace $\mathcal{U} \subseteq \mathcal{V}$ inherits a norm and inner product from $\mathcal{V}$.

The norm and inner product on $\mathcal{U}$ are the same as the norm and inner product on $\mathcal{V}$, just restricted to $\mathcal{U}$.

The subspace of symmetric matrices is particularly important in optimization and is given its own notation:

$$\mathbb{S}^n = \{X \in \mathbb{R}^{n \times n} : X^\top = X\}$$

Since it is a subspace of $\mathbb{R}^{n \times n}$, $\mathbb{S}^n$ inherits the Frobenius norm and trace inner product. (Except that people sometimes write the latter as $\langle X, Y \rangle = \text{tr}(XY)$. (Why?)).

Suppose we have two vector spaces $\mathbb{U}$ and $\mathbb{V}$, with norms $\|\cdot\|_{\mathbb{U}}$ and $\|\cdot\|_{\mathbb{V}}$, and inner products $\langle \cdot, \cdot \rangle_{\mathbb{U}}$ and $\langle \cdot, \cdot \rangle_{\mathbb{V}}$.

The **product space** $\mathbb{U} \times \mathbb{V}$ is the set of all ordered pairs (u, v) where $u \in \mathbb{U}$ and $v \in \mathbb{V}$.

Then a norm on $\mathbb{U} \times \mathbb{V}$ may be defined by

$$\|(u, v)\|_{\mathbb{U} \times \mathbb{V}} := \sqrt{\|u\|_{\mathbb{U}}^2 + \|v\|_{\mathbb{V}}^2}$$

An inner product on $\mathbb{U} \times \mathbb{V}$ is given by

$$\langle (u_1, v_1), (u_2, v_2) \rangle_{\mathbb{U} \times \mathbb{V}} := \langle u_1, u_2 \rangle_{\mathbb{U}} + \langle v_1, v_2 \rangle_{\mathbb{V}}$$

For example, an inner product on $\mathbb{R}^p \times \mathbb{R}^{m \times n}$ is given by

$$\langle (\mathbf{x}_1, Y_1), (\mathbf{x}_2, Y_2) \rangle = \mathbf{x}_1^\top \mathbf{x}_2 + \text{tr}(Y_1^\top Y_2)$$

and this induces the norm

$$\|(\mathbf{x}, Y)\| = \sqrt{\|\mathbf{x}\|_2^2 + \|Y\|_F^2}$$

Exercise

Write down Cauchy-Schwartz inequality for this inner product on $\mathbb{R}^p \times \mathbb{R}^{m \times n}$.

Convergence in $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$

Let $(X_k)_{k=1}^{\infty}$ be a sequence of matrices in $\mathbb{R}^{m \times n}$, i.e.,

$$X_k = \begin{bmatrix} x_{11}^{(k)} & \cdots & x_{1n}^{(k)} \\ \vdots & \ddots & \vdots \\ x_{m1}^{(k)} & \cdots & x_{mn}^{(k)} \end{bmatrix} \in \mathbb{R}^{m \times n}$$

for all $k \in \mathbb{N}$.

Note that, as usual, this includes as special case a sequence of vectors in $\mathbb{R}^m$ when we set $n = 1$ - in the following we will not distinguish the vector case unless necessary.

Definition (Convergence of matrices)

We say $\lim_{k \rightarrow \infty} X_k = X_*$ if

$$\lim_{k \rightarrow \infty} \|X_k - X_*\| = 0$$

That is, the norms of the errors is a convergent sequence in $\mathbb{R}$.

The definition of limit appears to depend on the choice of norms but actually it doesn't - if $\lim_{k \rightarrow \infty} \|X_k - X_*\|_a = 0$ in some norm $\|\cdot\|_a$, we must also have $\lim_{k \rightarrow \infty} \|X_k - X_*\|_b = 0$ for any other norm $\|\cdot\|_b$. This is because of the following.

Definition (Equivalent norms)

Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on $\mathbb{V}$ are **equivalent** if there exist constants $c_1, c_2 > 0$ such that

$$c_1\|v\|_a \leq \|v\|_b \leq c_2\|v\|_a$$

for all $v \in \mathbb{V}$.

Example of equivalent norms

Consider the two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on $\mathbb{R}^3$.

Let's consider the vector $\mathbf{x} = [1, -2, 3]$.

We have $\|\mathbf{x}\|_1 = |1| + |-2| + |3| = 6$ and $\|\mathbf{x}\|_2 = \sqrt{1^2 + (-2)^2 + 3^2} = \sqrt{14} \approx 3.74$.

We can find constants c_1 and c_2 such that

$$c_1 \|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq c_2 \|\mathbf{x}\|_1$$

We could choose $c_1 = 1$ and $c_2 = \frac{\sqrt{14}}{6}$ (for example) – this would work for these two vectors, but we need to find constants that work for all vectors.

For any vector $\mathbf{x} = [x_1, x_2, \dots, x_n]$, we can apply Cauchy-Schwarz to the vectors $[|x_1|, |x_2|, \dots, |x_n|]$ and $[1, 1, \dots, 1]$:

$$(|x_1| \cdot 1 + |x_2| \cdot 1 + \dots + |x_n| \cdot 1)^2 \leq (|x_1|^2 + |x_2|^2 + \dots + |x_n|^2) \cdot (1^2 + 1^2 + \dots + 1^2)$$

This simplifies to:

$$(|x_1| + |x_2| + \dots + |x_n|)^2 \leq (|x_1|^2 + |x_2|^2 + \dots + |x_n|^2) \cdot n$$

Therefore:

$$\|\mathbf{x}\|_1^2 \leq n \cdot \|\mathbf{x}\|_2^2$$

Taking the square root:

$$\|\mathbf{x}\|_1 \leq \sqrt{n} \cdot \|\mathbf{x}\|_2$$

Rearranging:

$$\|\mathbf{x}\|_2 \geq \frac{1}{\sqrt{n}} \cdot \|\mathbf{x}\|_1$$

This gives us $c_1 = \frac{1}{\sqrt{n}}$ (which is $\frac{1}{\sqrt{3}}$ for $\mathbb{R}^3$).

To find c_2 , we need to find the maximum value of $\|\mathbf{x}\|_2/\|\mathbf{x}\|_1$ across all vectors.

This ratio is maximized when only one component is non-zero. Consider $\mathbf{x} = [a, 0, 0]$: -
 $\|\mathbf{x}\|_1 = a + 0 + 0 = a$ - $\|\mathbf{x}\|_2 = a$

Therefore: $\|\mathbf{x}\|_2/\|\mathbf{x}\|_1 = |a|/|a| = 1$

In fact, from the triangle inequality, we can show that $\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$ for all $\mathbf{x} \in \mathbb{R}^3$ - so the ratio is always at most 1.

So the optimal constants for $\mathbb{R}^3$ are $c_1 = 1/\sqrt{3}$ and $c_2 = 1$.

All norms on finite dimensional vector spaces are equivalent

Theorem (All norms on finite dimensional vector spaces are equivalent)

All norms on finite dimensional vector spaces are equivalent.

This applies to spaces like $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$ because they are finite-dimensional but it fails in infinite dimensional spaces like $L^1(\mathbb{R})$ and $L^2(\mathbb{R})$.

Corollary

Convergence in $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$ does not depend on the choice of norm. The constants c_1 and c_2 for various pairs of norms can often be explicitly determined.

Exercise

Prove that for all $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \leq \sqrt{n}\|\mathbf{x}\|_2, \quad \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \sqrt{n}\|\mathbf{x}\|_\infty, \quad \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1 \leq n\|\mathbf{x}\|_\infty$$

Prove that for all $X \in \mathbb{R}^{m \times n}$,

$$\begin{aligned}\|X\|_2 &\leq \|X\|_F \leq \sqrt{\min(m, n)}\|X\|_2, \\ \frac{1}{\sqrt{n}}\|X\|_\infty &\leq \|X\|_2 \leq \sqrt{m}\|X\|_\infty, \\ \frac{1}{\sqrt{m}}\|X\|_1 &\leq \|X\|_2 \leq \sqrt{n}\|X\|_1.\end{aligned}$$

Example

If we have $\lim_{k \rightarrow \infty} \|\mathbf{x}_k - \mathbf{x}_*\|_1 = 0$, then the result in Exercise 4.1 shows that $\|\mathbf{x}_k - \mathbf{x}_*\|_2 \rightarrow 0$. Indeed, since

$$\|\mathbf{x}_k - \mathbf{x}_*\|_2 \leq \|\mathbf{x}_k - \mathbf{x}_*\|_1 \leq \sqrt{n} \|\mathbf{x}_k - \mathbf{x}_*\|_2 ,$$

by the squeezing lemma (or sandwich theorem), the middle term tends to 0 .
Conversely, if $\|\mathbf{x}_k - \mathbf{x}_*\|_1 \rightarrow 0$, then

$$\frac{1}{\sqrt{n}} \|\mathbf{x}_k - \mathbf{x}_*\|_1 \leq \|\mathbf{x}_k - \mathbf{x}_*\|_2 \leq \|\mathbf{x}_k - \mathbf{x}_*\|_1$$

so the middle term tends to zero. That is, convergence is independent of the choice of norm.

Theorem (Entry-wise convergence of matrices)

Let $(X_k)_{k=1}^{\infty}$ be a sequence of matrices in $\mathbb{R}^{m \times n}$. Then

$$X_k = \begin{bmatrix} x_{11}^{(k)} & \cdots & x_{1n}^{(k)} \\ \vdots & \ddots & \vdots \\ x_{m1}^{(k)} & \cdots & x_{mn}^{(k)} \end{bmatrix} \rightarrow \begin{bmatrix} x_{11}^* & \cdots & x_{1n}^* \\ \vdots & \ddots & \vdots \\ x_{m1}^* & \cdots & x_{mn}^* \end{bmatrix} = X_*$$

iff $x_{ij}^{(k)} \rightarrow x_{ij}^*$ for all $i = 1, \dots, m, j = 1, \dots, n$. That is,

$$\lim_{k \rightarrow \infty} \|X_k - X_*\| = 0 \iff \lim_{k \rightarrow \infty} x_{ij}^{(k)} = x_{ij}^*, \quad i = 1, \dots, m, j = 1, \dots, n$$

Proof: Use equivalence of $\|\cdot\|$ and $\|\cdot\|_{H,\infty}$.

We will discuss continuity of multivariate functions formally in the next lecture. For now, just take it to mean the usual thing: $\lim f(*) = f(\lim *)$, i.e., you can interchange the order $\lim$ and f .

Let $\lim_{k \rightarrow \infty} X_k = X_*$ and $\lim_{k \rightarrow \infty} Y_k = Y_*$. Then the following are continuous functions on $\mathbb{R}^{m \times n}$:

- Norm
- Inner product
- Linear combinations
- Matrix-matrix product
- Matrix inversion
- Trace
- Determinant

Norm is a continuous functions on $\mathbb{R}^{m \times n}$:

$$\lim_{k \rightarrow \infty} \|X_k\| = \|X_*\|.$$

Inner product is a continuous function on $\mathbb{R}^{m \times n} \times \mathbb{R}^{m \times n}$:

$$\lim_{k \rightarrow \infty} \text{tr} (X_k^\top Y_k) = \text{tr} (X_*^\top Y_*) .$$

Linear combination is a continuous function on $\mathbb{R}^{m \times n} \times \mathbb{R}^{m \times n}$:

$$\lim_{k \rightarrow \infty} \alpha X_k + \beta Y_k = \alpha X_* + \beta Y_*.$$

Matrix-matrix product is a continuous function on $\mathbb{R}^{m \times n} \times \mathbb{R}^{n \times p}$:

$$\lim_{k \rightarrow \infty} X_k Y_k = X_* Y_*$$

Matrix inversion is a continuous function on $GL(n)$ provided limit is in $GL(n)$: if X_k and X_* are invertible square matrices, then

$$\lim_{k \rightarrow \infty} X_k^{-1} = X_*^{-1}$$

We require X_* to also be invertible since it is possible for a sequence of invertible matrices to converge to a non-invertible matrix (e.g., a matrix with diagonal entries tending to zero).

Trace is a continuous function on $\mathbb{R}^{n \times n}$:

$$\lim_{k \rightarrow \infty} \operatorname{tr}(X_k) = \operatorname{tr}(X_*).$$

Determinant is a continuous function on $\mathbb{R}^{n \times n}$:

$$\lim_{k \rightarrow \infty} \det(X_k) = \det(X_*).$$

Again all these statements also apply to vectors.

In particular, the matrix-vector product is continuous.

We prove this as an example.

First note that by submultiplicativity $\|X\mathbf{y}\| \leq \|X\|\|\mathbf{y}\|$ for any $X \in \mathbb{R}^{m \times n}, \mathbf{y} \in \mathbb{R}^n$. So

$$\begin{aligned}\|X_k \mathbf{y}_k - X_* \mathbf{y}_*\| &= \|X_k \mathbf{y}_k - X_* \mathbf{y}_k + X_* \mathbf{y}_k - X_* \mathbf{y}_*\| \\ &\leq \|X_k \mathbf{y}_k - X_* \mathbf{y}_k\| + \|X_* \mathbf{y}_k - X_* \mathbf{y}_*\| \\ &\leq \|(X_k - X_*) \mathbf{y}_k\| + \|X_* (\mathbf{y}_k - \mathbf{y}_*)\| \\ &\leq \|X_k - X_*\| \|\mathbf{y}_k\| + \|X_*\| \|\mathbf{y}_k - \mathbf{y}_*\|\end{aligned}$$

and both the summands go to zero if $X_k \rightarrow X_*$ and $\mathbf{y}_k \rightarrow \mathbf{y}_*$. All statements in **prp-4p16** may be similarly proved.

Other notions like Cauchy sequences, series, subsequences, etc, generalize to sequences of vectors and matrices verbatim.

Cauchy sequences

Definition of a Cauchy sequence

Cauchy sequences are a generalization of the concept of convergent sequences to higher-dimensional spaces.

In a Cauchy sequence, the terms get arbitrarily close to each other as the sequence progresses.

It tells us that a sequence is “converging” even if we don’t know its limit yet.

Definition (Cauchy sequence)

A sequence $(X_k)_{k=1}^{\infty}$ is **Cauchy** if for all $\varepsilon > 0$ there is some $N \in \mathbb{N}$ such that $\|X_k - X_j\| < \varepsilon$ whenever $k, j > N$.

Theorem (Cauchy sequences are convergent)

Every Cauchy sequence in $\mathbb{R}^{m \times n}$ is convergent, and converges to a limit in $\mathbb{R}^{m \times n}$.

We have a specific name for a property where every Cauchy sequence converges to a limit in the space: we say that the space is **complete**.

So the space $\mathbb{R}^{m \times n}$, with any norm, is complete.

We write this as $(\mathbb{R}^{m \times n}, \|\cdot\|)$ is complete.

This is a very important property, and we will use it extensively in future lectures.

Definition (Compact sets)

A subset $S \subseteq \mathbb{R}^{m \times n}$ is **compact** if every open cover for S has a finite subcover.

See next slide for an easier way to visualize this!

In $\mathbb{R}^{m \times n}$, compact sets are exactly the closed and bounded sets.

Theorem (Heine-Borel theorem)

A subset S of $\mathbb{R}^{m \times n}$ is compact if and only if S is closed and bounded.

This means that a set is compact if and only if it

- contains all its boundary points,
- is contained in some finite ball.

Those two properties will be easy to check!

In a compact set, any infinite sequence must have a subsequence that converges to a point in the set.

Theorem (Bolzano-Weierstrass theorem)

Let $S \subseteq \mathbb{R}^{m \times n}$. Then S is compact iff every sequence in S has a convergent subsequence with limit in S .

Note that because our set is compact, we know that any convergent sequence will converge to a point in the set.

What this theorem adds is that there must *always* be at least one convergent subsequence in every infinite sequence.

Theorem (Matrix geometric series)

If $X \in \mathbb{R}^{n \times n}$ and $\|X\| < 1$, then

$$\sum_{i=0}^{\infty} X^i = (I - X)^{-1}$$

In particular, the series converges, and the limit is invertible.

Proof of matrix geometric series theorem

We first want to show that the series converges.

Before we do that, though, let's review a result with real numbers (instead of matrices like we have here).

Suppose we have a scalar α with $0 < \alpha < 1$, and we consider the infinite sum

$$\sum_{i=0}^{\infty} \alpha^i$$

Does this sum converge?

Yes, it does! (In fact, it converges to $\frac{1}{1-\alpha}$.)

But we can check the convergence by seeing if the sequence of partial sums is Cauchy – that is, if adjacent partial sums get closer and closer to each other.

Let's write down the difference between the k th and j th partial sums. Assume that $k > j$.

$$S_k - S_j = \sum_{i=0}^k \alpha^i - \sum_{i=0}^j \alpha^i = \alpha^{j+1} + \alpha^{j+2} + \dots + \alpha^k$$

$$S_k - S_j = \sum_{i=0}^k \alpha^i - \sum_{i=0}^j \alpha^i = \alpha^{j+1} + \alpha^{j+2} + \dots + \alpha^k$$

The partial sum of a geometric series is given by

$$\sum_{i=j+1}^k \alpha^i = \alpha^{j+1} \frac{1 - \alpha^{k-j}}{1 - \alpha} \leq \alpha^{j+1} \frac{1}{1 - \alpha}$$

Since $\alpha < 1$, for any $\epsilon > 0$, we can find some N such that if $j > N$, then $\alpha^{j+1} \frac{1}{1-\alpha} < \epsilon$. The sequence of partial sums is Cauchy, and hence the series converges.

Now we can use this result to show that the series of matrices converges.

Intuitively, we still need to show that S_k and S_{k+1} get closer and closer to each other for large k .

Since S_k is a sum of a k matrices, and S_{k+1} is a sum of a $k + 1$ matrices, the difference between them is just the last matrix in S_{k+1} .

We need to show that this matrix gets smaller and smaller as k gets larger.

Let $\alpha = \|X\|$. We were told that $\|X\| < 1$, so $\alpha < 1$. We will also assume that $\alpha > 0$ (otherwise X is the zero matrix, and the series is just 0).

For $k \geq 0$, write $S_k = \sum_{i=0}^k X^i$. Then

$$\|S_k - S_j\| = \left\| \sum_{i=j+1}^k X^i \right\| \leq \sum_{i=j+1}^k \|X\|^i = \sum_{i=j+1}^k \alpha^i = \alpha^{j+1} \frac{1 - \alpha^{k-j}}{1 - \alpha} \leq \alpha^{j+1} \frac{1}{1 - \alpha}$$

Now we are just where we were with the real numbers – $\alpha^{j+1} \frac{1}{1 - \alpha}$ can be made arbitrarily small by choosing j large enough.

Thus the sequence of partial sums is Cauchy, and hence the series converges.

Next, we need to show that $S = \lim_{k \rightarrow \infty} S_k$ the inverse of $I - X$.

Note that $\lim_{k \rightarrow \infty} X^k = 0$ since

$$\|X^k - 0\| = \|X^k\| \leq \|X\|^k = \alpha^k \rightarrow 0$$

The next step multiplies the partial sum by $(I - X)$.

When we expand this product, most terms cancel out, leaving only $I - X^{k+1}$.

This is like what happens when you multiply $(1 + x + x^2 + \cdots + x^k)$ by $(1 - x)$ - you get $1 - x^{k+1}$.

$$S_k(I - X) = \left(\sum_{i=0}^k X^i \right) (I - X) = I - X^{k+1}.$$

Letting $k \rightarrow \infty$,

$$S(I - X) = \lim_{k \rightarrow \infty} S_k(I - X) = \lim_{k \rightarrow \infty} (I - X^{k+1}) = I$$

Hence $S = (I - X)^{-1}$, as desired.

- We talked about vector spaces and linear maps. Nothing new, just reviewing the basic definitions.
- We introduced the idea of norms, and discussed the two norms we'll use often: the vector 2-norm and the Frobenius norm.
- We reviewed the idea of the inner product, and talked about how it can induce norms.

Then we talked about how to bring the idea of convergence to vector spaces.

- We defined Cauchy sequences, and said that they are convergent in finite-dimensional vector spaces.
- We introduced the idea of compactness, and showed how to use the Heine-Borel theorem to check if a set is compact.
- Finally, we used Cauchy sequences to show that the matrix geometric series converges, and hence we can write the inverse of $I - X$ as an infinite series.